Does the Age at Exposure to Violence Matter?

Evidence from the end of paramilitary violence in Colombia

Job market paper proposal

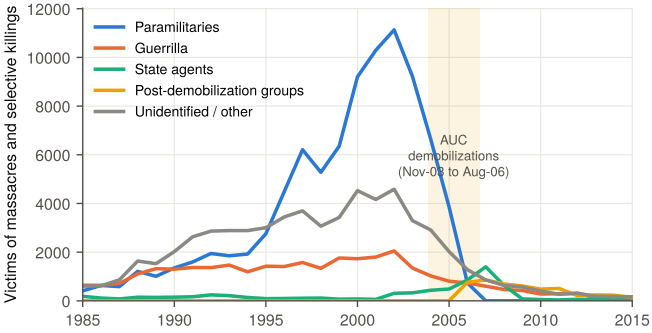
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What is the question?

Early-life shocks have different effects depending on the age at which they hit. Does this hold for violence, and through which channel?

- Sensitive periods in skill formation: Cunha and Heckman (2007, *AER*); Almond and Currie (2011, *JEP*); Carneiro, Salvanes, Willage and Willén (forthcoming, *EJ*).
- Childhood exposure to violence lowers adult human capital: León (2012, *JHR*); Duque (2024, *JCR*); Akbulut-Yuksel, Tekin and Turan (2025, *JHR*).
- What is not known: the **age gradient**. Is exposure at 0–2 worse than at 3–5 or at school age? Does the damage run through the caregiver or through the child's own experience?
- Why it is hard: it requires a shock that *starts and ends* on known dates, and children located *before* the shock. Without both, the profile is contaminated: the replication of León by Huaroto, Gonzales and Pacheco (2026, *JHR*) finds "effects" of violence that occurred before conception.

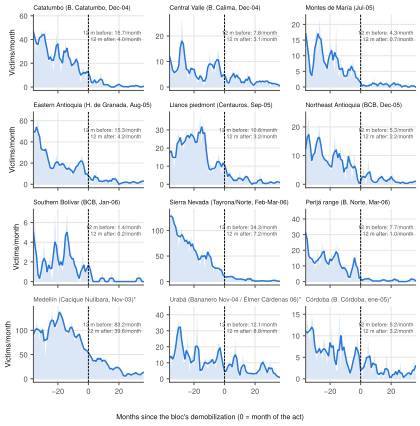
Why Colombia? Paramilitary violence has an actor and an end



- Paramilitary expansion 1997–2002; peak 2001–02.
- Ceasefire December 2002; the 38 blocs demobilize between November 2003 and August 2006.
- Paramilitary victims are zero from 2007: the end is real.
- Guerrilla violence continues, and the victim registry identifies the perpetrator.

CNMH, Observatorio de Memoria y Conflicto: victims of massacres and selective killings by perpetrator.

Why Colombia? The end is dated municipality by municipality



* Zones whose post period is contaminated by post-demobilization groups. Series: victims of massacres and selective killings attributed to paramilitaries and post-demobilization groups (CNMH-OMC, March 2025 cut); 3-month moving average.

- 451 municipalities account for 94% of paramilitary victims 1997–2006; median presence of 9 years.
- Each bloc leaves on a different date; violence in its zone falls in the months before and stays at zero after.
- For each municipality m we date the start S_m and the end T_m of paramilitary presence.
- **Source of variation:** children born in the same municipality were of different ages when the violence ended. Born in 2003 in Montes de María: never exposed; born in 1998: exposed from age 0 to 7.

What is the challenge? Households move

Colombia has one of the largest conflict-driven internal migrations in the world: in affected municipalities the median share of population displaced in 1997–2010 is 23%.

- Exposure depends on whether the household stays and when it leaves; leaving is itself a different treatment.
- Whoever is observed in the municipality years later is not who was there: locating people at the outcome mixes the effect with selection.

Our answer.

- Locate children **before** the shock: exposure is assigned by *municipality of birth*, in individual microdata. The estimand is intention to treat: the effect of being born where violence will end at age a, household response included.
- Displacement is measured, not assumed: date and reason of the last move (GEIH) and displaced population by municipality and year (RUV).

How do we identify the effect?

Person i born in municipality m in year c, observed in survey year t:

$$Y_{imct} = \sum_a \beta_a D^a_{mc} + \gamma \, \text{Post}_{mc} + \varphi \, G_{mc} + \mu_m + \lambda_{c \times d(m)} + \theta_t + \varepsilon_{imct}$$

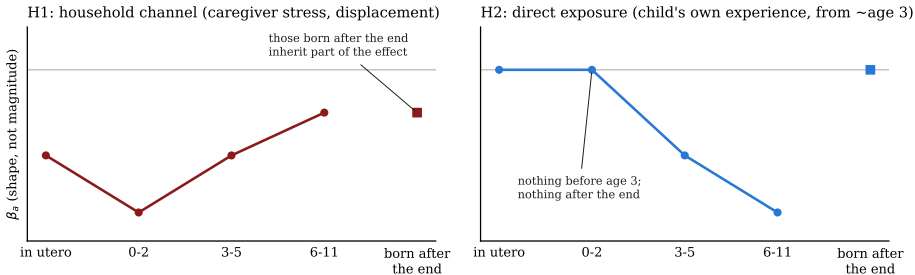
- $a \in \{\text{in utero}, 0\text{–}2, 3\text{–}5, 6\text{–}11\}$. D^a_{mc}: years of paramilitary presence in m (between S_m and T_m) that cohort c lived through at age a; zero in untreated municipalities and at ages after T_m. Post_{mc}: born after T_m in a treated municipality.
- G_{mc}: years of guerrilla presence at the same ages. μ_m: municipality of birth. $\lambda_{c \times d(m)}$: cohort $\times$ department. θ_t: survey year and age at survey. Standard errors clustered by municipality of birth (451 treated, 670 comparison).
- β_a **is the effect of one year of paramilitary violence lived at age** a, comparing cohorts born in the same municipality who were of different ages when the violence ended, net of department-level cohort trends. The variation comes from the timing of T_m (and S_m) within departments.
- Outcomes: years of schooling, secondary completion, tertiary enrollment, employment, formality, labor earnings.

Which data?

Source	Content	Role
GEIH 2012–2024 (DANE household survey)	Municipality of birth, age, schooling, employment, formality, earnings, last migration ($\sim$0.75 M persons per year; cohorts 1988–2003 observed at ages 20–36)	Outcomes and location before the shock
CNMH-OMC victim registry	Victims by municipality, month and perpetrator, 1985–2015	S_m, T_m, D^a_{mc}, G_{mc}
RUV (victims' registry)	Displaced population by municipality and year	Displacement channel
Vital statistics (EEVV)	Births 1998–2010 and deaths 1998–2019 by municipality	Selection into birth and survival

In hand: CNMH and RUV, with S_m and T_m built for the 451 municipalities. Public, to be downloaded: GEIH and EEVV. Extension if access is granted: ICFES linked records (displacement, health, formal employment at the individual level).

What do we expect, and what would it tell us?



- **H1, household channel:** the damage runs through the caregiver (stress, displacement, income loss): it starts in utero, peaks at 0–2, and those born *after* the end inherit part of it ($\gamma < 0$).
- **H2, direct exposure:** the child must be old enough to encode the experience: nothing before age 3, rising damage with age, nothing for those born after the end ($\gamma = 0$).
- Second step: β_a separately for households that were displaced and those that stayed.

What could go wrong?

Threat	Test
Cohorts in treated municipalities were already on a different path	Placebo: violence assigned to the months before conception must have zero effect; Rambachan and Roth (2023, *REStud*)
Violence changes who is born and who survives	Births and deaths by municipality and cohort (EEVV); Lee (2009, *REStud*) bounds by age window
Guerrilla and state violence in the same places	G_{mc} and state-agent violence at the same ages; the guerrilla profile as a perpetrator placebo
Continuous, staggered treatment	de Chaisemartin and D'Haultfœuille (2024, *REStud*); Callaway, Goodman-Bacon and Sant'Anna (2024); event study around T_m

What is the contribution?

1. **The age gradient of a violence shock, from an episode that ends.** First estimates separating in utero, 0–2, 3–5 and 6–11, with the end of exposure dated by municipality and location fixed before the shock. The dated end is what makes the comparison clean: cohorts born in the same municipality before and after the violence stopped.
2. **A mechanism test:** household channel versus direct exposure, from the shape of the profile and from those born after the end; then by household displacement.
3. **Adult outcomes:** schooling and labor market at ages 20–36, not only school test scores.

Relative to Duque (2024, *JCR*): a different decade, an actor with a dated end, municipality of birth instead of residence, adult outcomes. Relative to León (2012, *JHR*) and its replication: the pre-conception placebo is part of the design, not an afterthought.

Timeline. Months 1–2: data and power. Months 3–5: main estimates, placebos, displacement split. Month 6: first draft and go/no-go. Months 7–18: robustness and full paper.